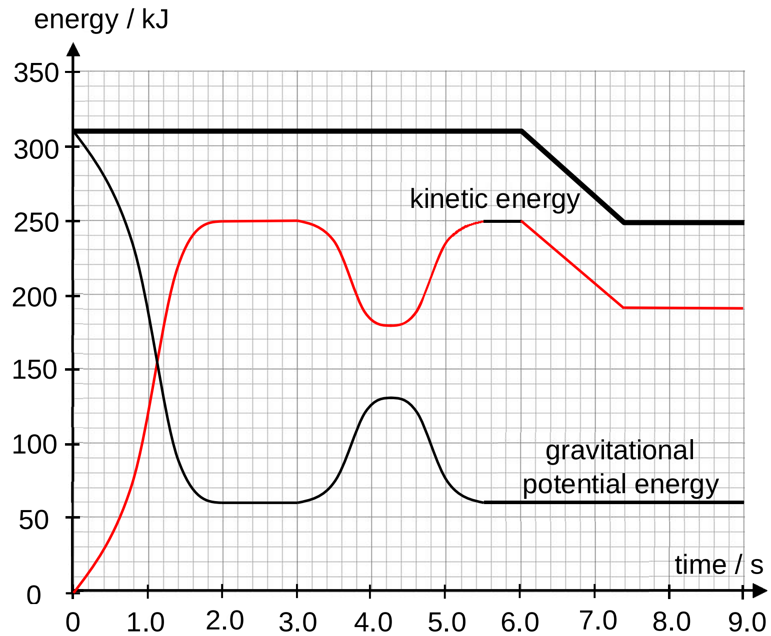
			= 1.7 N [A1]	
4	**(a)**	**(i)**	 [1] rise from *t* = 0 s to *t* = 6.0 s till K.E is 250 kJ [CC: What abt the shape of the curve before KE 250kJ and dip in KE between t = 3s and 5.4s?] [1] straight line joining points (6.0 , 250) and (7.4 , 190), followed by horizontal line from *t* = 7.4 s to *t* = 9.0 s, with mechanical energy at 190 KJ	

4	(a)	(ii)	From Fig 7.3, t = 3.0 s to t = 4.3 s, the increase in gravitational potential energy is $\Delta GPE = 130 - 60 = 70$ kJ $mg\Delta h = 70000$ $\Delta h = \dfrac{70000}{380 \times 9.81}$ $= 18.78 \approx 19$ m The increase in height, and therefore the diameter d of loop, is 19 m.	 M1 A1
4	(a)	(iii)	$F_c = \dfrac{mv^2}{r}$ $= \dfrac{2(K.E)_{top}}{\frac{d}{2}}$ $= \dfrac{2(180000)}{\frac{18.78}{2}}$ N + mg = 38339 N = 34611 = 34600 N	 M1 A1
4	(b)	(i)	rate of loss of KE $= \dfrac{\text{loss of KE}}{\text{duration in zone A}}$ $= \dfrac{(250 - 190) \times 10^3}{7.4 - 6.0}$ $= 42857$ W ≈ 43000 W	 M1 A1
4	(b)	(ii)	$P = Fv$ $F = \dfrac{P}{v}$ $= \dfrac{42\ 857}{34.0}$ $= 1260$ N	 M1 A1
5	(a)		V_R = 4.8 (3.5)(0.80) [M1]	